



Determinant Factors of Fintech Adoption in Organization using UTAUT Theory Approach

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ABSTRACT

The presence of Financial Technology (FinTech) as an innovative and disruptive financial service can improve the efficiency and scope of organization business process through the application of technology. In determining the effectiveness of Fintech adoption, each organization will measure by considering both from the perspectives of management and user in the organization. This proposed study aims to examine the effect of FinTech adoption in organization using the UTAUT approach. It attempts to measure the variable of performance expectancy, effort expectancy, social influence, customer's trust and regulatory services into the effect of FinTech adoption in the organization. This study adopts quantitative method through online survey and the data will be analyzed by using Structural Equation Method (SEM). This proposed study's findings could contribute by enhancing the organization the importance of FinTech adoption to increase business performances. The study showed that the the adoption of FinTech in organization had significantly influenced by performance expectancy, effort expectancy, social influence, customer trust as well as regulatory services. Customer trust itself had the highest influenced among others. Meanwhile, regulatory services had the smallest influence to the FinTech adoption in Organization.

Introduction

Covid-19 pandemic contributed to the rise of digital payments as people stayed at home and no longer able to visit stores and offices, switching to e-commerce. The ease of access and the advancement of technology information enforce the organization to change the business model as the response of new digital transformation especially in the financial industries. Financial Technology (FinTech) services as an innovative technology enabler provide digitized financial services in simpler and convenient way for the organization (Lee & Shin, [2018](#)). The usage of FinTech in the organization was expected to minimize the transaction cost as well as increase the financial services (Blohm & Leimeister, [2013](#)). Asia has been the winner

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of mobile payments for decades. A study conducted by (Kurniasari, [2021](#)) has found that the Asia Pacific region reached the highest market share of digital payments in the world, with 44% point of sale transactions using mobile phones and 69% online shop transactions were settled through a mobile device.

During the Covid-19 pandemic, the total FinTech market in Indonesia reached Rp 128.7 trillion (Kurniasari, [2021](#)). Internet literacy index among people at the age of 15 years old+ in Indonesia has grown tremendously and reached 95.7% of the total population. It is predicted that by 2025, there will be 233.03 million Indonesians who are internet users while 226.47 million are mobile internet users. Internet banking still has become favorite digital payment method. Electronic money circulation grew at the 60% level annually and showed an upward trend with 68% increase in nominal transactions in the third quarter of 2022 compared with the same quarter in the previous year (Indonesia, [2022](#)).

Even more Indonesian are open-minded toward FinTech usage, many still have doubts in settling their company's financial transaction through advanced technological devices (Kurniasari, [2018](#)). Therefore, the usage of FinTech especially in the organization is not well-utilized (Venkatesh & Bala, [2009](#)). While digital payment providers are offering promotion and attractive services, the real number rate of the digital payment usage in the organization did not fulfill the target that was established by the government (Kurniasari, [2022](#)). Many organization employees and management are still reluctant to use digital payment for settling the business financial transactions. They are still choosing to use conventional ways such as: bank transfers or checked payment since the business transactions usually involve huge amounts of money.

Using the Unified Theory of Acceptance and Use of Technology (UTAUT) approach, the variables of performance expectancy, effort expectancy, social influence and trust had a significant effect on customer adoption of the new FinTech platform (Zhang, [2017](#)). In addition, since financial industries are strongly related with the highly regulated business, the government policy in terms of regulatory services was the main consideration for the consumers in adopting FinTech digital payments.

Therefore, the research objective is to analyze the decision FinTech adoption in organization to support the business process activities. The purpose of this research is to analyze which variable of the UTAUT model with the additional variable of regulatory services that had the highest effect in influencing the adoption of digital payment services in organization. The findings of the research will make some recommendations for FinTech stakeholders in making a supportive regulation that can accommodate customer's interest in conducting business financial transactions and improving business performances.



Literature Review

Financial Technology (FinTech)

Financial technology (FinTech) is often associated as a mobile based technology system with its function to increase efficiency and effectiveness of financial performance (Kim, [2015](#)). The new financial digital platform changed the organization business model by adding some new attractive values for its customers (Walchek, [2020](#)). Information technology changed business life since all relationships and networking were interconnected with the technology (Lim, [2020](#)). The birth of FinTech was triggered by the new sharing economy system model and the government regulation support as the response to the advancement of information technology (Blohm & Leimeister, [2013](#)).

Performance Expectancy

Performance expectancy is related to the people's belief level that using the new FinTech digital payment will assist them to increase their business performance (Venkatesh., [2003](#)). Performance expectancy was proven to have a positive influence in the adoption in the new internet banking services (Daka, [2019](#)). Many researchers associated the performance expectancy as a main construct with the theory of perceived usefulness that was previously introduced by Davis, ([2009](#)). In their study of FinTech adoption in the banking industry, performance expectancy was explained by the financial transaction accessibility that was related with the simplicity and easiness in using the new FinTech digital platform (Dwivedi, [2021](#)).

Effort Expectancy

Users are willing to accept and use the technology only if they feel easy and less effort in using all the available features. Effort expectancy clearly described as the easiness level in accessing the new technology platform (Daka, [2019](#)). In the research in the banking industry in the UEA, effort expectancy was associated with the level of digital accessibility that supports the organization performance (Dwivedi, [2021](#)).

Social influence

Social influence is related with the level to which an individual perceives the other's effect as a direct determinant (Venkatesh., [2003](#)) and usually has a significant impact on the use of FinTech services in the banking industry (Kurniasari, [2020](#)). Indonesia is well-known as a communal country, social relationships among close-knit families and friends have influence the decision to adopt new technology platforms (Hofstede, [2022](#)). This type of relationship and the group networking were able to initiate a social capital source of funds that was easily accessed among the



inner circle (Dwivedi, [2021](#)). In addition, the advancement of FinTech made all the networking relationships and transactions conducted virtually (Venkatesh., [2003](#)).

Customer Trust

Trust is a key determinant factor for the users in adopting and using the new technology platform (Lee, [2019](#)). Trust is dependent on the relationship quality that is developed among the parties and related with the aspect of reliability and trustworthiness (Arpaci, [2018](#)). The strong relationship quality created if the FinTech provider was able to provide necessary and accurate information as well as provided the supported employee to handle customer's problems (Malaquias, [2020](#)). If the customers had a secure (Kurniasari, [2022](#)), safety and confidence (Schierz, [2020](#)) toward the reliability of the FinTech in settling the business transaction, they are willing to adopt this new system in the organization (Li, [2019](#)).

Regulatory Services

Government was the main actor in the development of the FinTech ecosystem in Indonesia by releasing related regulations to protect both the customers and industries interest (Dwivedi, [2021](#)). Goo & Heo, ([2020](#)) explained that the active role of government was needed to reduce uncertainty and risks in adopting the new FinTech platform (Liebana-Cabanillas, [2018](#)). The risk can be minimized if there are some legal regulations to ensure safety in using the digital platform. Mwiya, ([2017](#)) stated that users will not be interested in adopting higher risk platforms with uncertain regulation.

Financial Technology Adoption

The FinTech adoption decision was dependent on the interest and preference in using this digital payment platform for business transaction purposes (Kurniasari, [2022](#)). Since this research was conducted in the context of organization, the business profile was also the key determinant in creating the adoption behaviour of FinTech (Goo & Heo, [2020](#)). Once the user feels comfortable and familiar in using the new technology for the business transaction, they will decide to use it. Board of management in the organization become the key player in ensuring that all the organization business process aligned the FinTech adoption strategy (Dwivedi, [2021](#)) by providing necessary training (Glavina, [2020](#)) and communication flow (Panchal, 2019) to increase the capabilities in using FinTech.

Based on the proposed framework, the study was being able to develop some hypotheses as follows:

H₁: Performance Expectancy has a positive effect on FinTech Adoption in the Organization.



H₂: Effort Expectancy has a positive effect on FinTech Adoption in the Organization.

H₃: Social Influence has a positive effect on FinTech Adoption in the Organization

H₄: Customer Trust has a positive influence on FinTech Adoption in the Organization.

H₅: Regulatory Services has a positive influence on has a positive effect on FinTech Adoption in the Organization.

Method

The Unified Theory of Acceptance and Use of Technology (UTAUT) concept was used to conduct this research along with the additional concept of consumers' trust and regulatory services. As a quantitative research with hypothesis testing, this study will find the influence of the UTAUT variable as well as trust and regulatory services variable toward the FinTech adoption in the organization. The study collected the data through an online survey. There's 186 respondents who're able to answer 5-point Likert scale questions based on the scientific indicators for each variable (Nunnally, 1978). The indicators for UTAUT variables taken from previous study initiated by (Venkatesh., 2003). Meanwhile, the indicators for regulatory services adopted from the research of (Goo & Heo, 2020). All the primary data further tested statistically using Structural Equation Model with Lisrel 8.80 software. Following are the details of indicators for each variable:

Table 1
The Indicators of Each Variable

Variables	Dimensions	Instruments
PERFORMANCE EXPECTANCY (Independent Variable)	• Financial Transaction Accessibility (X1)	X1.1 Using the Fintech system allow me to perform tasks quickly X1.2 I use the system to expand my business X1.3 I don't have any difficulties in using the Fintech
EFFORT EXPECTANCY (Independent Variable)	• Digital Accessibility (X2)	X2.1 The Fintech platform is useful to support my work X2.2 My interaction with the system is clear and understandable X2.3 Fintech is simple and easy to use
SOCIAL INFLUENCE (Independent Variable)	• Group Networking (X3)	X3.1 I am joining in social media group X3.2 I always communicating with



Variables	Dimensions	Instruments
Variable)		other member of the group X3.3 I am doing online business transaction frequently
CUSTOMER TRUST (Independent Variable)	Relationship Quality (X4)	X4.1 I need adequate information before doing my financial transaction X4.2 I always need staff support in doing my financial transaction
REGULATORY SERVICES (Independent Variable)	Customer Protection (X5)	X5.1 I am feeling secured when I am borrowing the money from the Bank X5.2 I know that the government make a clear regulation to protect the customer X5.3 The financial digital platform is not regulated clearly by the government X5.4 I know there's a formal government body if I have some complaint about the financial institution services
	Industry Protection (X6)	X6.1 There must be a government regulation to control the financial digital protection X6.2 The government must set-up maximum limit of digital transaction
FINTECH ADOPTION IN ORGANIZATION (Dependent Variable)	Customer Decision (Y1)	Y1.1 I interested in adopting Fintech for business transaction Y1.2 I prefer to use Fintech platform in settling the business transaction
	Business Profile (Y2)	Y2.1 My employee is more than 50 persons Y2.2 I always using Fintech in running my business Y2.3 I am doing Fintech for business transaction

Result and Discussion

Demographic data explained the characteristic of the respondents; 59.5% respondents were female with the majority age being 36-45 years old (50%) and 60.2% respondents hold Bachelor Degree. 60% of respondents had a job position as Senior Staff/Department Head with the average working experience length 4-5 years



(45%). 68% respondents currently worked as employees in the private companies from various industries.

Using Pearson correlation method and pre-test, it found that the critical value of $t > 0.871$, showed that all indicators had strong correlation. Reliability test using Cronbach's Alpha coefficient showed the number of 0.911, which means that this research is valid. To measure the fitness of the research model, the Confirmatory Factor Analysis (CFA) was used. Analysis toward the structured model was used to see the correlation toward the latent variable. The table 2 showed that all GOF indicators are at the good fit which means that there were perfect theories in this research. The table 3 showed the value of R Square and Adjusted R Square for each variable:

Table 2
Design Summary for Goodness for Fit Testing Model

GOF Indicator	Estimated Value	Testing Result	Conclusion
<i>Absolute Fit Value</i>			
GFI	$GFI \geq 0.90$	0.92	Good Fit
RMSEA	$RMSEA < 0.08$	0.03	Good Fit
<i>Incremental Fit Value</i>			
NNFI	$NNFI > 0.90$	0.98	Good Fit
NFI	$NFI > 0.90$	0.94	Good Fit
AGFI	$AGFI > 0.90$	0.93	Good Fit
RFI	$RFI > 0.90$	0.92	Good Fit
IFI	$IFI > 0.90$	0.91	Good Fit

Source: Data Analysis using LISREL 8.80

Table 3
R-Square and Adjusted R-Square

Variables	R-Square	Adjusted R-Square
Adoption to FinTech	0.862	0.956
Performance Expectancy	0.598	0.910
Effort Expectancy	0.628	0.927
Social Influence	0.515	0.898
Customer Trust	0.651	0.936
Regulatory Services	0.492	0.891

Source: Data Analysis using LISREL 8.80

Table 4
Result of Hypothesis Testing

Hypotheses	Variables	Coefficient Standard	t-Value	Statistical Conclusion
H ₁	PE → FA	0.000	6.692	Data Supported
H ₂	EE → FA	0.000	6.820	Data Supported



Hypotheses	Variables	Coefficient Standard	t- Value	Statistical Conclusion
H ₃	SI → FA	0.005	4.836	Data Supported
H ₄	CT → FA	0.000	8.725	Data Supported
H ₅	RS → FA	0.046	4.832	Data Supported

The structural equation model of this research was described as follow:

$$FA = 0.000*PE + 0.000*EE + 0.005*SI + 0.000*CT + 0.000*RS, \text{ Errorvar.} = 0.39, R^2 = 0.956$$

$$(0.040) \quad (0.050) \quad (0.033) \quad (0.068) \quad (0.029)$$

$$6.692 \quad 6.820 \quad 4.836 \quad 8.725 \quad 4.832$$

Following is the explanation of the result of hypotheses testing for each variable:

H₁: Performance Expectancy has a positive effect on FinTech Adoption in the Organization, with the t-value = 6.692 and the effect value of 0.000. The finding of the research indicated that performance expectancy has a positive influence on the adoption of FinTech in organization. The finding is supported by Wiradinata, (2018); Alwi, (2019); Chong, (2019) and Kurniasari, (2022), who claim that consumers tend to adopt new FinTech platform services that can allow them to perform tasks quickly. Kurniasari, (2021) explains that the usefulness of FinTech in organization can expand the businesses and lead to better organization performances.

H₂: Effort Expectancy has a positive effect on FinTech Adoption in the Organization, with the t-value = 6.820 and the effect value of 0.000. The research analysis reveals that effort expectancy has proven to have significant influence over the adoption of FinTech platform services in the organization. Consistent results can be found from studies conducted by Dwivedi, (2021); Liebana-Cabanillas, (2018) and Aji, (2020). Findings of this study indicate that most of the respondents who are employees at the private companies consider the ease of use when using applications as main consideration in their decision to adopt FinTech services in the organization. The FinTech platform is useful to support their works in the organization since the features of FinTech itself are simple, clear, easy to understand and reduce human error barriers Kurniasari, (2022).

H₃: Social Influence has a positive effect on FinTech Adoption in the Organization, with the t-value = 4.836 and the effect value of 0.005. The findings of the research show that the employees were easily influenced by their group networking within the organization to adopt the use of FinTech. People are aware of others' opinions in their social circle when deciding to adopt the new technology. In Indonesia, the higher level of management requires obedience and will convince the subordinates to adopt FinTech for the organization's interest since it will bring greater business performances (Panchal, 2019).



H4: Customer Trust had a positive effect on FinTech Adoption in the Organization with the t-value of 8.725 and the effect value of 0.000. The user paid greater attention to all the relationship quality that was developed by the availability of information before doing the financial transaction. The finding was supported to by research Chen, (2014) who explained that trust has a positive effect before users make decisions in adopting new technology. Employees put a high priority on trust when it comes to utilizing the new system and ensure that there will be full support if they find any difficulties or problems. Kurniasari, (2018) claims that employees are reluctant in using FinTech services to handle business transactions because they have uncertainty regarding the privacy concerns, information technology and institution image and reputation.

H5: Regulatory Services had a positive effect on FinTech Adoption in the Organization with the t-value of 8.725 and the effect value of 0.000. This finding supports the previous study conducted by Lee & Shin, (2018); Malaquias, (2020) and Chen, (2014). The result of the research showed that the users believed that the government was a main actor in controlling the FinTech ecosystem and would give protection both for customers and industries. Users are willing to adopt the new FinTech platform as long as they believe that the government has a clear regulation and policy to manage any financial risk including the security issues and data privacy Malaquias, (2020) and Chen, (2014). The clear regulation will give guidance to conduct the FinTech business transparently and ensure that the new secure platform would keep the secrecy of its users (Lee, 2019).

Conclusion

All the variables used in this research such as performance expectancy, effort expectancy, social influence, customer trust and regulatory services were proven to have a positive influence toward the FinTech adoption in organization. Customer trust had the highest effect in influencing the employee decision to adopt the FinTech platform in the organization, meanwhile regulatory services had the lowest influence. The implication of the research shows that organizations had to develop employee trust so they are willing to adopt the new FinTech system. The role of management support was needed to ensure that the employees are able to use the system smoothly in settling their business transactions.

This finding was expected to contribute in reducing the sceptic behaviour of the organization in adopting and using the new platform to support the business performances. It also supported the Indonesian government to become more advanced in financial business and reach financial inclusion across the country.

The limitation of this study was its only use of quantitative methods without using other qualitative methods to confirm and strengthen the analysis.



The study showed that the higher R-square result achieved 95.6%. This means that only 4.4% the adoption of FinTech payment services in organization was influenced by other factors. The further research should also consider the national culture (Hofstede, 2022) as well as ICT infrastructure (Malaquias, 2020) as the factors that can influence the adoption level of FinTech in the organization. There's an urgency to develop a collaboration within the FinTech ecosystem to make each stakeholder adopt and use the new FinTech platform without any doubt. The FinTech provider should deliver the highest standard of service quality and security protection to attain the organization competitiveness and performance (Schierz, 2020).

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